

My journey to Expert on Numerai-signals*

Sandro Scodeller

based in Oslo, Norway

April '24

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*Or so I thought

Who am I?
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Intermezzo
Next steps
“Reaching the top” or so it seemed
Status now

Layout

1 My Background

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- Everyone sees strong finance background, right?

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- I bring to you from the far north of Norway: Niuton



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Niuton in a shell

- Derives variables from a sosio-dynamic mesh modelling the movement in the stock-market using Newtonian mechanics.

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- Emerging markets → reduce overhead of US or Europe investments
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- we have pilots running cf. <https://www.niuton.io/>

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What is the Problem?



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What is the Problem?



Is a solution indeed, but sub-optimal

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Niuton in a shell

- Niuton is comprised of 3 main systems:
 - Niuton Core

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Niuton in a shell

- Niuton is comprised of 3 main systems:
 - Niuton Core
 - Stability system

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 - Niuton Core
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- There are in total 58x2 base-parameters
- These should be optimized, but not all of same importance
- These optimizations different for specific markets → Numerai is just another market.

Lesson learned #1

The above statement is a over-simplification

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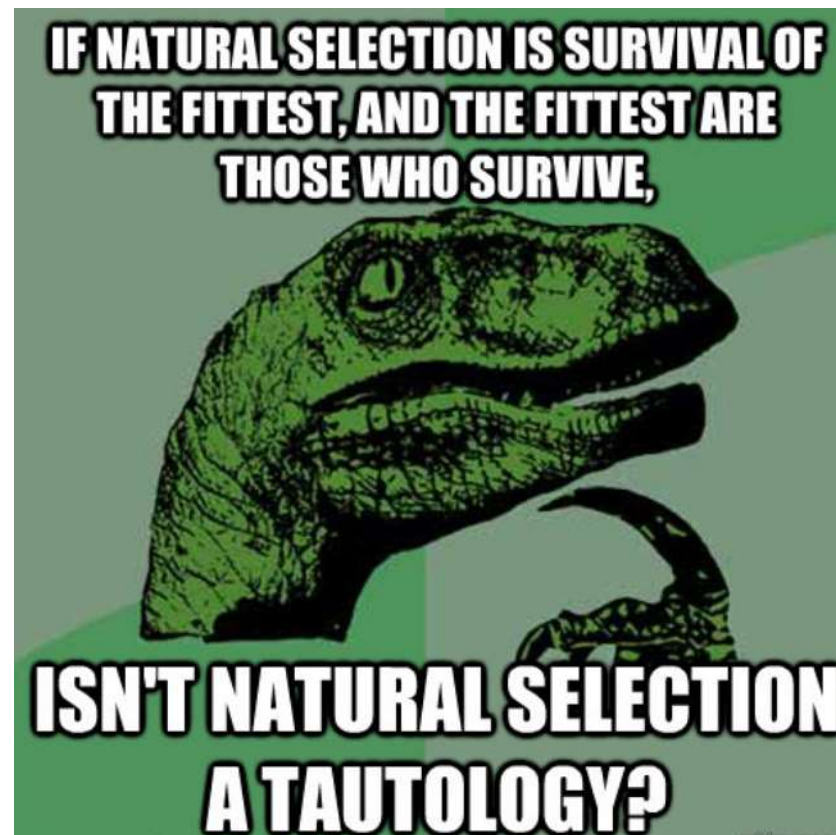
Genetic algorithm

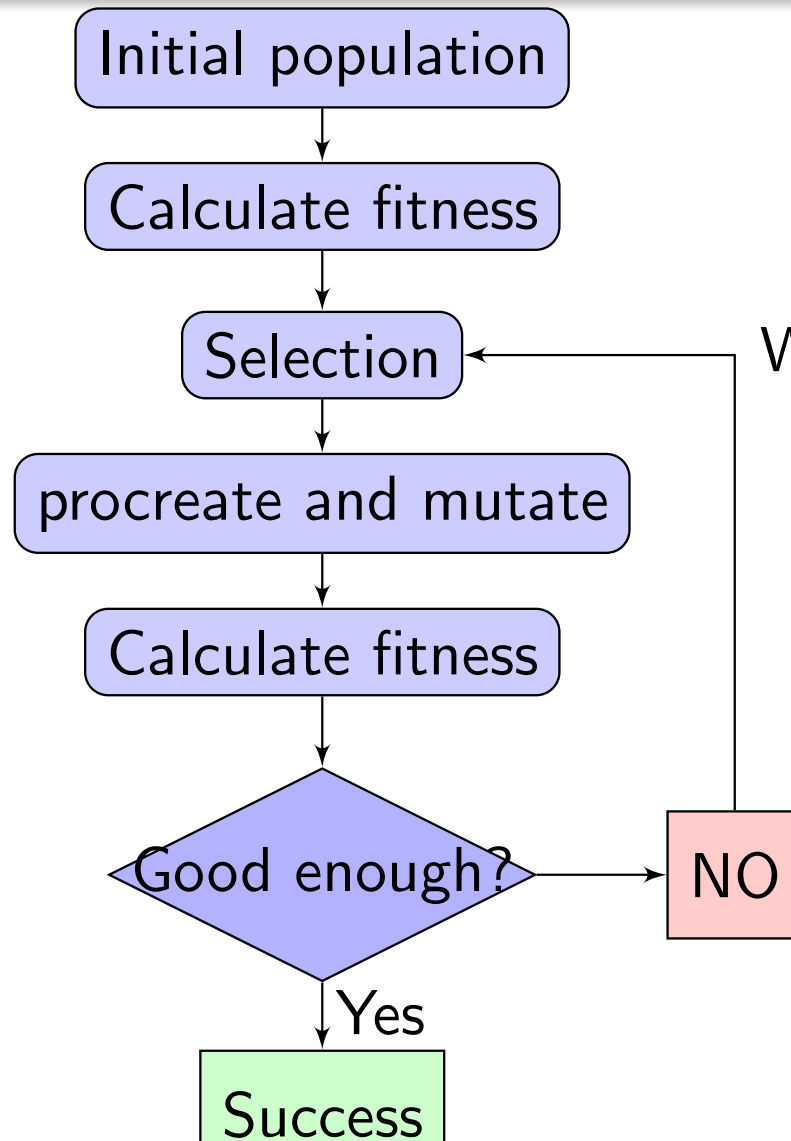
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Genetic algorithm

Genetic algorithm





Example [1,2], [3,2], [4,4], [1,3], [2,0]

Who can procreate? How does it work?

Elitism

[Elite], offspring

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We have it all?

- Figure out how to define 1 signal from 58 sweetspots.

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We have it all?

- Figure out how to define 1 signal from 58 sweetspots.
- Just read the docs and

Lesson learned #2

Understanding the docs is part of the competition, finding info too... A quiz, based on info from docs: what is actually a signal?

Lesson learned #2

Understanding the docs is part of the competition, finding info too... A quiz, based on info from docs: what is actually a signal?

From docs all you USED get that for instance P/E-ratio is a signal, just saw it was improved

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- ~~Just~~ read the docs and deploy some models - genetically optimized for different markets

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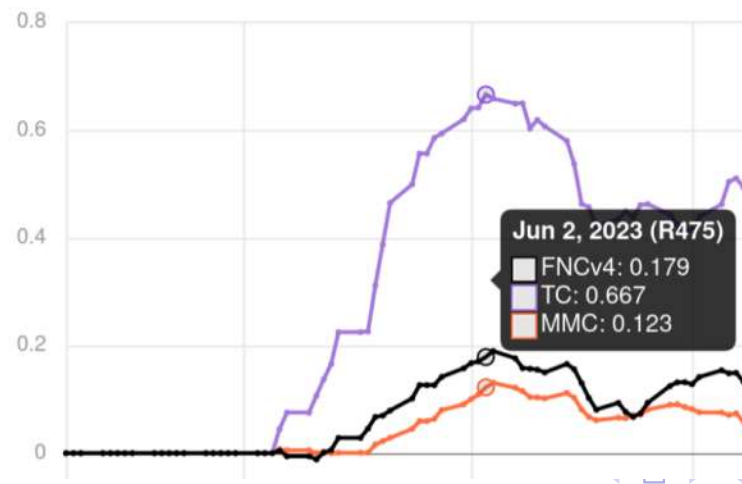
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- Had these available: Tokyo, N8000, SP500, Crypto (Binance-only), Oslo and also make a forest from 58

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- **Just** read the docs and deploy some models - genetically optimized for different markets
- Had these available: Tokyo, N8000, SP500, Crypto (Binance-only), Oslo and also make a forest from 58
- Half expected to now just sit back and watch:



Lesson learned #3

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Bahamas is a bit further away :-)

- Had a remote machine available - Windows... :-)

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• Use task-scheduler to run my python-files

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• Both weekly training and daily submissions

- Had a remote machine available - Windows... :-(
• Use task-scheduler to run my python-files
• Fetch stock-market data from twelvedata.com and compute the Niuton variables for all Numerai-stocks.
• Both weekly training and daily submissions
• Soon reaching limit.

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- 3 months in all looks good:

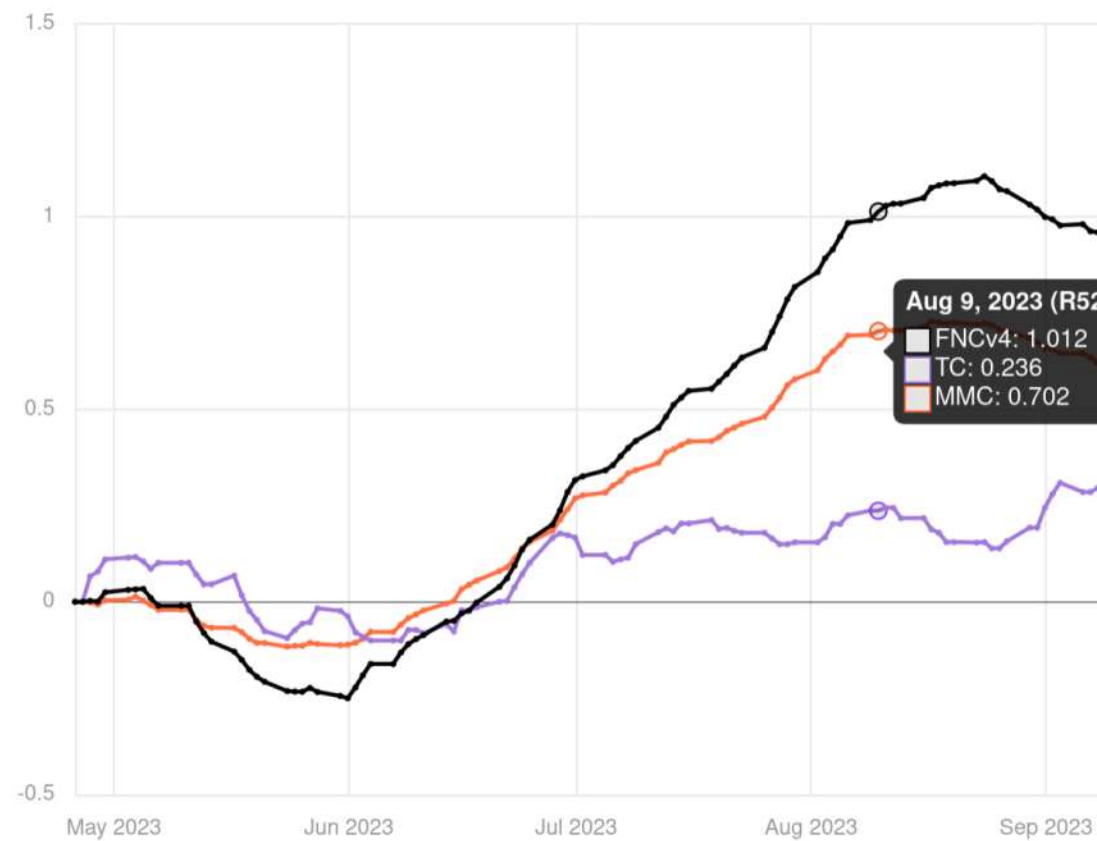
Return

Cumulative stake-weighted-average returns of all models in the past 1 year



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- naive Sandro thinks just fluctuations - she'll be alright..

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- naive Sandro thinks just fluctuations - she'll be alright..
- for a while it fluctuates, then just goes down (never re-trained, no surprise now)

Lesson learned #4

Lesson learned #4

You have to react as early as you notice, because unstake happens only after round-completion

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- Proceed to build models (random forests)

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- What timerange to train on? What are all the targets?:evil-start: let's maybe check the docs :evil-end:

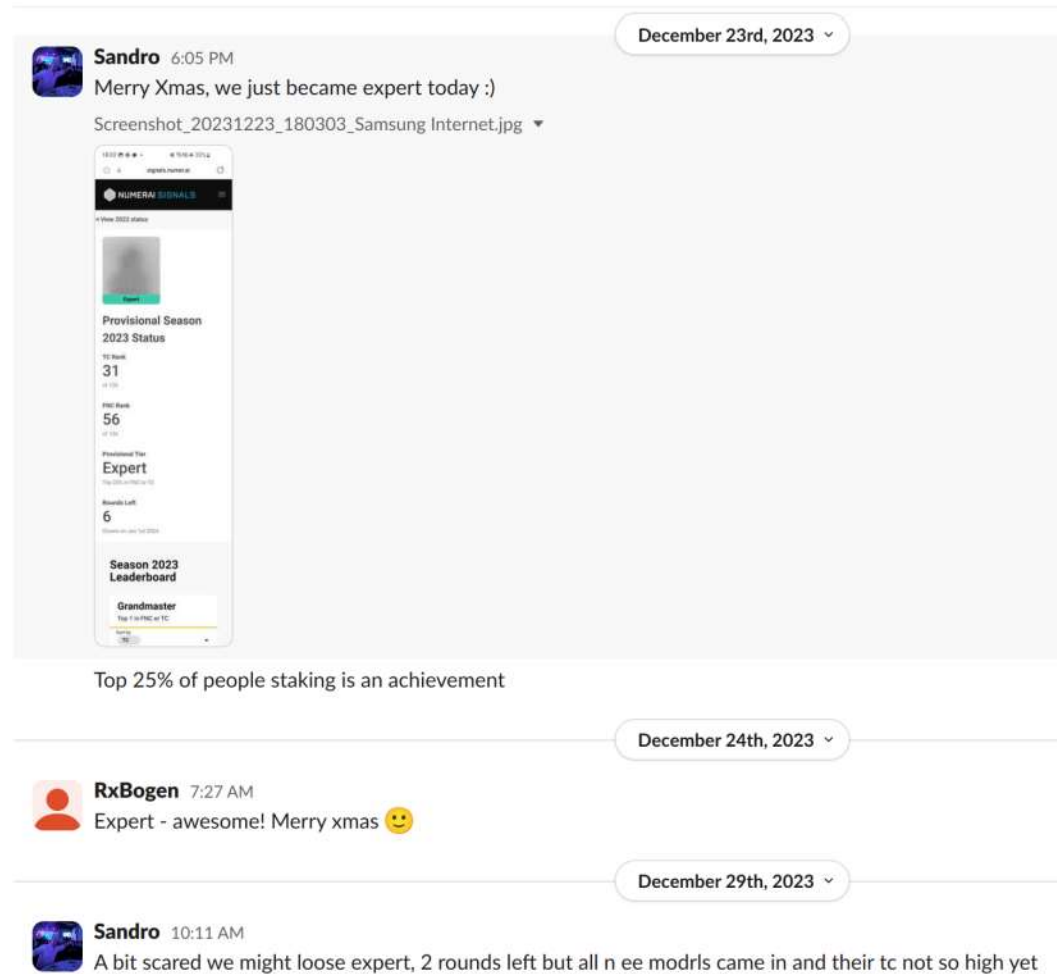
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- first deploy models models combining 2 targets trained on 4months of data, later also a monthly and a weekly version

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- first deploy models models combining 2 targets trained on 4months of data, later also a monthly and a weekly version
- this gets me back here:



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Lesson learned #5

Lesson learned #5

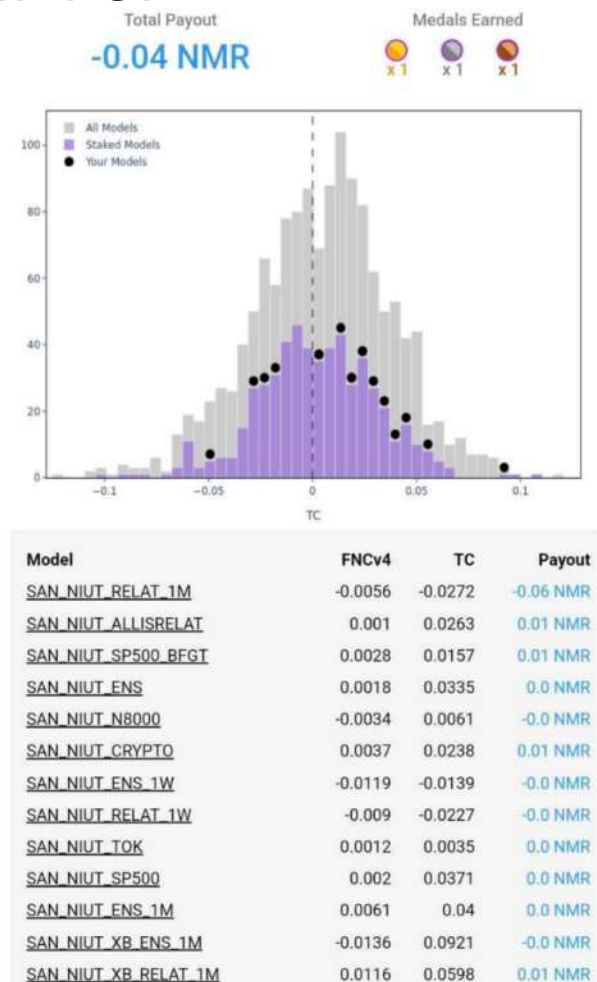
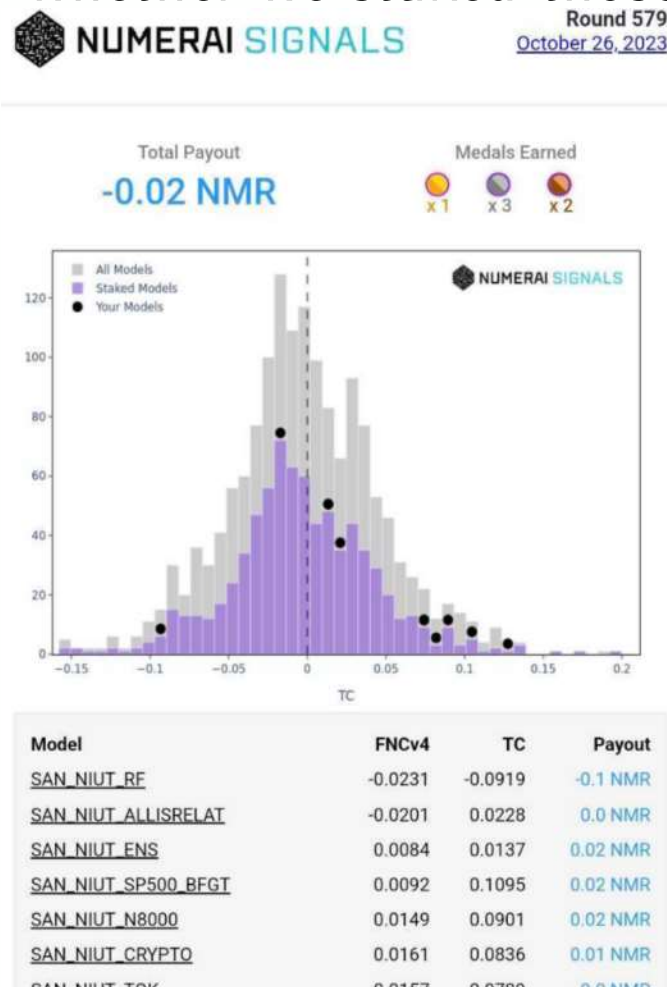
It finishes when the rounds complete

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- There is more: we got gold medals on TC

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- There is more: we got gold medals on TC
- Guess whether we staked those on TC?



Lesson learned #6

Lesson learned #6

Staking is its own science

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- Somewhere along the way I hear XGBoost is amazing

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- some are, but not all(!) I tried (submitting first ones in December)

- Somewhere along the way I hear XGBoost is amazing
- some are, but not all(!) I tried (submitting first ones in December)
- kiss goodbye to Expert

Lesson learned #7

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If you care about your position on the leaderboard do not try new things in December

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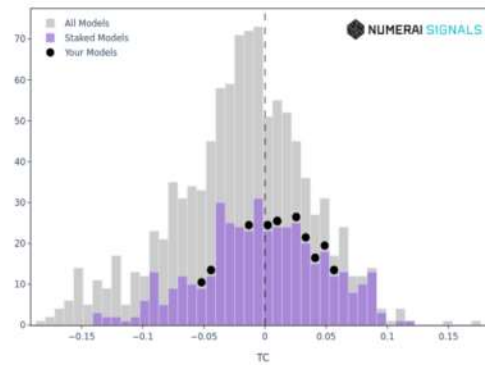
- Numerai sets new fix stake-multipliers

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- Looked good for most models, the best ones we increase the stake

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- Looked good for most models, the best ones we increase the stake
- see Lesson learned #6 or here:

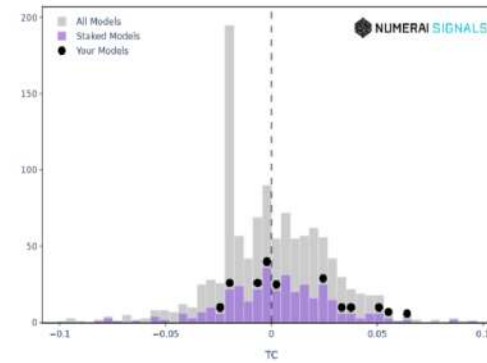
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SWA TC 0.0016
 SWA FNCv4 -0.0155
 PAYOUT -0.1127 NMR



Model	TC	FNCv4	At Risk	Payout
<u>SAN_NIUT_RELAT_1M</u>	-0.0434	-0.0219	3.79 NMR	-0.19 NMR
<u>SAN_NIUT_TOK</u>	0.0153	-0.0276	2.41 NMR	0.0035 NMR
<u>SAN_NIUT_SP500_BFGT</u>	-0.0375	-0.031	2.39 NMR	-0.12 NMR
<u>SAN_NIUT_ENS_1M</u>	0.0486	0.0155	2.28 NMR	0.1142 NMR
<u>SAN_NIUT_SP500</u>	-0.0093	-0.0279	1.31 NMR	-0.03 NMR
<u>SAN_NIUT_ALLISRELAT</u>	0.0381	0.0026	1.26 NMR	0.0478 NMR
<u>SAN_NIUT_ENS</u>	0.0283	0.0001	1.04 NMR	0.0283 NMR
<u>SAN_NIUT_N8000</u>	0.0145	-0.0388	0.9 NMR	-0.0 NMR
<u>SAN_NIUT_CRYPTO</u>	0.0146	-0.0297	0.58 NMR	-0.0 NMR
<u>SAN_NIUT_ENS_1W</u>	0.0636	0.0007	0.53 NMR	0.0265 NMR
<u>SAN_NIUT_RELAT_1W</u>	-0.0455	-0.0193	0.49 NMR	-0.02 NMR
<u>SAN_NIUT_XB_RELAT_1W</u>	0.0623	0.0005	0.22 NMR	0.0109 NMR
<u>SAN_NIUT_XB_ENS_1M</u>	0.0045	0.0105	0.21 NMR	0.002 NMR
<u>SAN_NIUT_XB_ENS_1W</u>	0.047	0.0084	0.2 NMR	0.0098 NMR
<u>SAN_NIUT_XB_RELAT_1M</u>	0.0067	0.0037	0.19 NMR	0.0016 NMR
<u>SAN_NIUT_XB_ENS_4M</u>	0.0273	0.0031	0.16 NMR	0.0045 NMR
<u>SAN_NIUT_XB_RELAT_4M</u>	0.0497	-0.0207	0.14 NMR	0.0055 NMR
<u>SAN_NIUT_OSE</u>	0.0153	-0.0276	0.01 NMR	0.0 NMR

SWA TC 0.01
 SWA FNCv4 -0.0051
 PAYOUT -0.0214 NMR



Model	TC	FNCv4	At Risk	Payout
<u>SAN_NIUT_ENS_1M</u>	-0.0063	-0.0161	2.92 NMR	-0.08 NMR
<u>SAN_NIUT_RELAT_1M</u>	0.0022	0.0003	2.34 NMR	0.011 NMR
<u>SAN_NIUT_ALLISRELAT</u>	0.0556	0.0114	1.92 NMR	0.0961 NMR
<u>SAN_NIUT_SP500_BFGT</u>	0.0004	-0.0184	1.72 NMR	-0.03 NMR
<u>SAN_NIUT_TOK</u>	-0.0201	-0.0009	1.17 NMR	-0.05 NMR
<u>SAN_NIUT_ENS</u>	0.0009	-0.0155	1.12 NMR	-0.02 NMR
<u>SAN_NIUT_SP500</u>	-0.0025	-0.0134	0.85 NMR	-0.02 NMR
<u>SAN_NIUT_ENS_1W</u>	0.0677	-0.0089	0.82 NMR	0.041 NMR
<u>SAN_NIUT_N8000</u>	-0.0201	-0.0035	0.62 NMR	-0.03 NMR
<u>SAN_NIUT_RELAT_1W</u>	0.0029	0.0131	0.4 NMR	0.0076 NMR
<u>SAN_NIUT_CRYPTO</u>	-0.0174	0.0055	0.28 NMR	-0.01 NMR
<u>SAN_NIUT_XB_RELAT_1W</u>	0.0359	0.0082	0.23 NMR	0.0113 NMR
<u>SAN_NIUT_XB_RELAT_1M</u>	0.0511	0.0237	0.22 NMR	0.011 NMR
<u>SAN_NIUT_XB_ENS_4M</u>	0.0527	0.0148	0.2 NMR	0.0102 NMR
<u>SAN_NIUT_XB_ENS_1W</u>	0.0284	-0.0013	0.19 NMR	0.0097 NMR
<u>SAN_NIUT_XB_RELAT_4M</u>	0.0383	0.0074	0.13 NMR	0.0067 NMR
<u>SAN_NIUT_XB_ENS_1M</u>	0.0416	-0.0065	0.07 NMR	0.0034 NMR
<u>SAN_NIUT_OSE</u>	-0.0201	-0.0009	0.01 NMR	-0.0 NMR

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Weeding out

- Made choices on MMC stake-multiplier crossing fingers.

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Weeding out

- Made choices on MMC stake-multiplier crossing fingers.
- Biggest lesson learned of all

Lesson learned #8

Lesson learned #8

Discord is not the thing you see on top right of the numerai website and ignore thinking you'll find all in the docs and forum!

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- Thanks to Discord I can now run diagnostics, so less finger crossing

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- Can come here to learn

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- Can come here to learn
- can ask for useful things like returning number of submitted and accepted tickers -(for 5months I did not submit US-ticks for example due to own bug)

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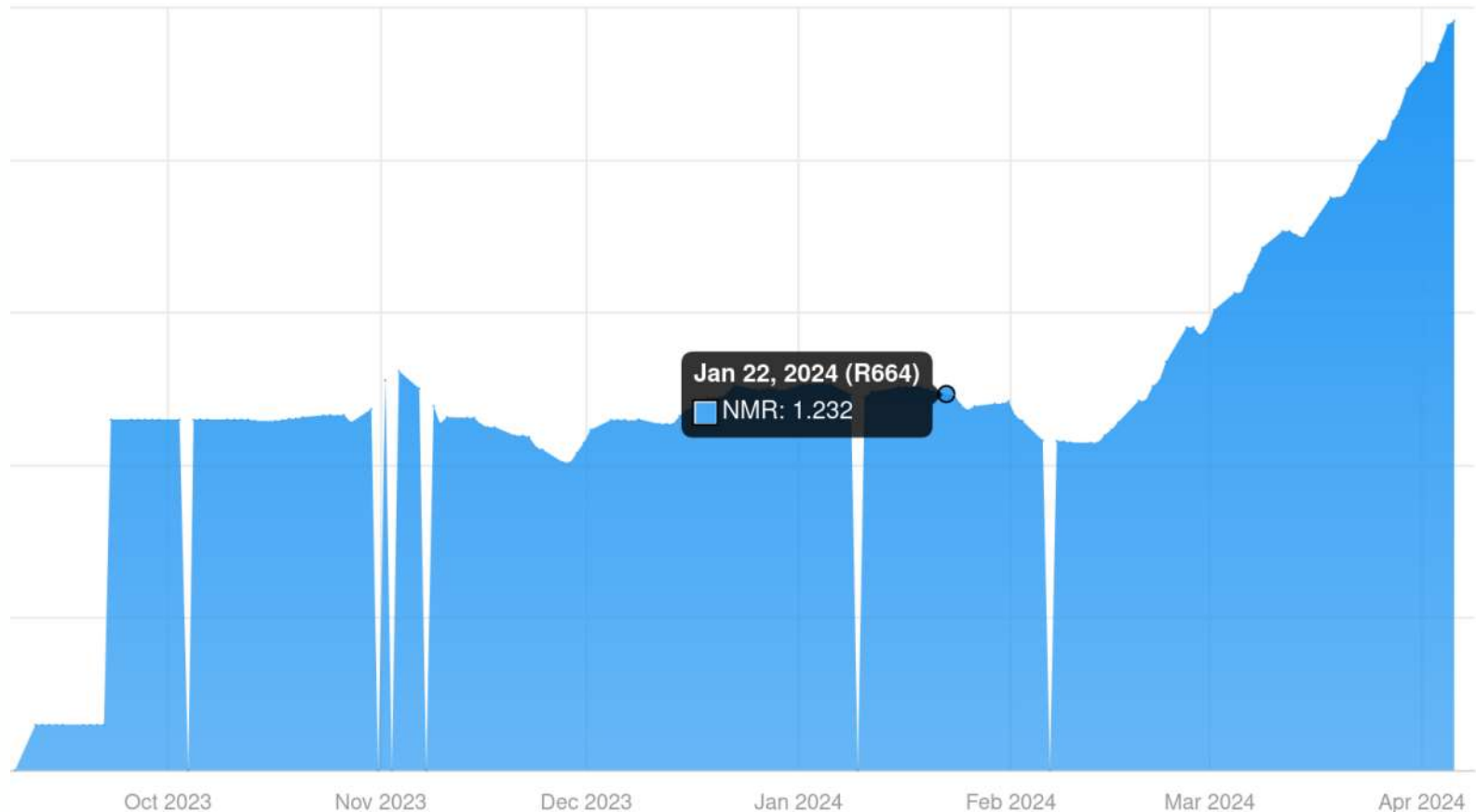
Runar



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Doing now

- being happy with the recent changes:



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	1Y Avg	Rank
TC	0.0022	633
FNCV4	0.0022	281
CORRV4	0.0019	279
ICV2	0.0029	478
RIC	0.0036	291
MMC	0.0026	425

Current Model Correlations

CWSNMM	0.0318
MCWSM	0.0449
APCWSM	-0.0005

1 Day Return	▼ 2.9%
3 Month Return	▲ 98.8%
1 Year Return	—
All Time Return	▲ 132.3%
Multipliers	1×FNCv4 2×TC
Meta Model Control	0.0061%

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SAN_N		CORRV4	ICV2	RIC	FNCV4	MMC	STAKE	1D	▼ 3M	1Y
20	 SAN_NIUT_ALLISRELAT	0.0025	0.0016	0.0040	0.0029	0.0030	2.902 NMR	0.53%	141.3%	—
56	 SAN_NIUT_XB_RELAT_1M	0.0027	0.0041	0.0030	0.0029	0.0020	0.32 NMR	4.5%	73.4%	—
59	 SAN_NIUT_XB_RELAT_1W	0.0014	0.0008	0.0024	0.0013	0.0016	0.37 NMR	9.1%	71.2%	—
61	 SAN_NIUT_XB_RELAT_4M	0.0021	0.0018	0.0026	0.0023	0.0019	0.23 NMR	1.2%	70.3%	—

- What is relat? Taken from forum making all features and target relative , ranking per day

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- Not original idea, but our variables are (Niuton)

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- Not original idea, but our variables are (Niuton)
- More hours → amazing so: any investors here ?:-)

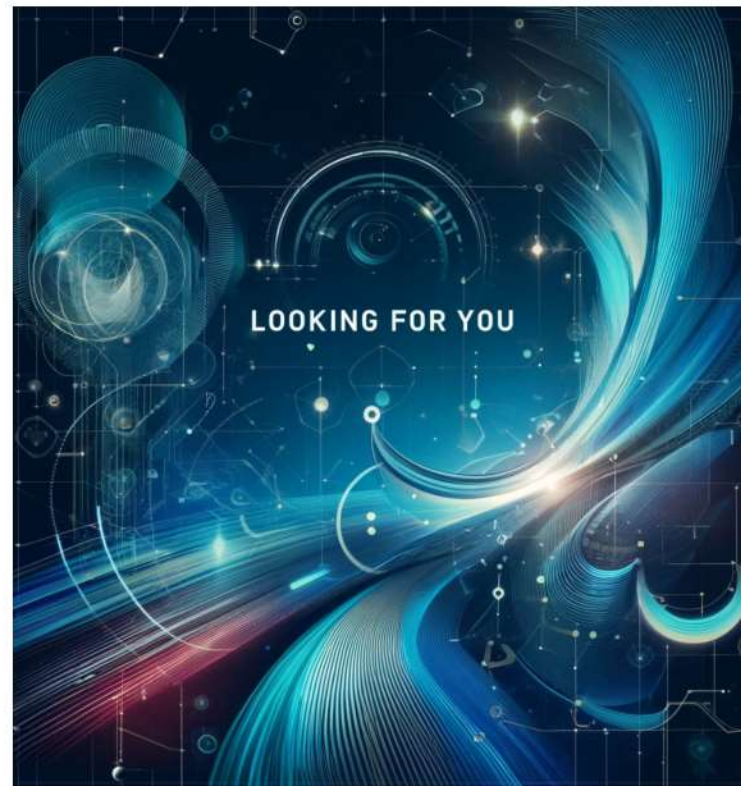
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- Simple deal: revenue-share

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- Simple deal: revenue-share
- EVENLY!

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- Simple deal: revenue-share
- EVENLY!
- Niuton puts the stake

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- Talk to me after or reach out to sandro@niuton.io

- Simple deal: revenue-share
- EVENLY!
- Niuton puts the stake
- Talk to me after or reach out to sandro@niuton.io
- Let's make this even greater.

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**IT WILL BE
FUN!**
THEY SAID.
lessons learned the hardway